Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT(Data Science) Sem-II Regular/Remedial April-June 2024

U72A1IP2: INTRODUCTION TO PROGRAMMING-II

Time: 3 Hours Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any Five)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Write a short note on Two-Dimensional Array.	06	CO1	BTL3
2	What is Character Array? Explain Declaration and Initialization of String Variables.	06	CO1	BTL5
3	Discuss Elements of user defined function.	06	CO2	BTL3
4	What is the difference between call by value and call by reference? Discuss it with example.	06	CO4	BTL4
5	Discuss fopen() and fclose() function with example.	06	CO4	BTL5
6	How to copy and compare structure variable? Discuss it with example.	06	CO3	BTL6
7	Write a short note on DMA.	06	CO5	BTL3
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Discuss run time initialization of one dimensional array with example.	06	CO1	BTL6
2	List out various category of UDF. Discuss any one category with example.	06	CO2	BTL5
3	What is Recursion? Explain it with proper example	06	CO2	BTL6
Q.3	**Answer the following: (Any Two)**	**12**		
1	Write a short note on pointer and structure.	06	CO4	BTL3
2	Discuss array within structure with example.	06	CO3	BTL6
3	How to use command line argument in file I/O operation ? Discuss it.	06	CO4	BTL5
Q.4	**Answer the following: (Any One)**	**06**		
1	Discuss Common programming Errors.	06	CO5	BTL3
2	List out Macro substitution and Explain any one in detail.	06	CO5	BTL5

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B. Sc.IT (Data Science) Sem-II Reg/Rem April-June 2024

U72A2DA2: DATA ANALYSIS-II

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Define Statistical functions with examples.	06		BTL-1
2	Define Database functions with examples.	06		BTL-1
3	List various types of charts, Discuss any two charts in detail.	06		BTL-2
4	Describe the process of combining data from multiple sources and analyzing alternative data sets.	06		BTL-1
5	List steps for analyzing data dynamically using PivotTables.	06		BTL-1
6	Explain how to include documents in a workbook and discuss the benefits of including Documents in a Workbook.	06		BTL-2
7	Discuss Following Terms: Header, Footer, Hyperlink, Page Breaks	06		BTL-2
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Define Financial functions with examples.	06		BTL-1
2	Describe how to summarize data using Sparklines and create dynamic charts using Pivot Charts.	06		BTL-2
3	Explain how to customize the appearance of charts.	06		BTL-2
Q.3	**Answer the following: (Any Two)**	**12**		
1	List the advantages of using PivotTables for summarizing and analyzing large datasets.	06		BTL-1
2	Describe the process of filtering, showing, and hiding PivotTable data.	06		BTL-2
3	What is macro? Write down steps to create and run macro in Excel.	06		BTL-2

Q.4	**Answer the following: (Any One)**	**06**		
1	Discuss different setting options while printing a workbook or worksheet.	06		BTL-2
2	Analyze the impact of changing page breaks on the layout and formatting of a worksheet.	06		BTL-5

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B. Sc.IT (Data Science) Sem-II Regular/Remedial April-June 2024

U72A3DMS: DATABASE MANAGEMENT SYSTEM

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	What is database? Simplify advantages of DBMS in details.	06	CO1	BTL2/BTL4
2	Elaborate Three Levels of the Architecture in details.	06	CO1	BTL5
3	What is normalization? Analyze Boyce Codd normal form (BCNF) in details.	06	CO2	BTL2/BTL4
4	Evaluate pattern matching operator in detail.	06	CO3	BTL5
5	Classify the DML SQL command in details.	06	CO3	BTL4
6	Evaluate joins in SQL in details.	06	CO4	BTL5
7	Demonstrate view and sequence in details.	06	CO5	BTL3
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Analyze the responsibility of Database Administration.	06	CO1	BTL4
2	What is E-R diagram? Explain relationship of E-R-diagram in details.	06	CO2	BTL2/BTL4
3	Apply Functional Dependency with example.	06	CO2	BTL3
Q.3	**Answer the following: (Any Two)**	**12**		
1	What are constraints? Categorize foreign key and check constraints in details.	06	CO3	BTL2/BTL4
2	Apply string Functions with example.	06	CO4	BTL3
3	Illustrate the group by and having clause in SQL.	06	CO4	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	Evaluate Grant and revoke feature with example.	06	CO5	BTL5
2	Design normalizes database design with ER relationship of Library Management System.	06	CO5	BTL6

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT (Data Science) Sem-II Regular April-June 2024

U72A4COA COMPUTER ORGANIZATION AND ARCHITECTURE

Time: 3 Hours Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Describe the role of data representation in computer systems.	06	CO1	BTL-4
2	Multiplication Using Simple Method. a. (10101)2*(11)2 b. (11111)2*(101)2 c. (1001001)2*(110)2	06	CO1	BTL-5
3	Explain Pin diagram of 8085 microprocessor.	06	CO4	BTL-4
4	Define parallel execution and instruction execution. How do they differ in terms of their goals and mechanisms?	06	CO3	BTL-3
5	Describe Microcontroller and Its Application.	06	CO2	BTL-4
6	What is an addressing format? Explain the differences between direct addressing mode and indirect addressing mode.	06	CO3	BTL-5
7	Discuss the significance of input/output (I/O) instructions in Assembly Language programming. How do I/O instructions such as IN and OUT facilitate communication between the CPU and external devices?	06	CO5	BTL-6
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Explain the difference between ASCII and EBCDIC.	06	CO2	BTL-4
2	What is Error? Explain types of error detection method.	06	CO1	BTL-4
3	Differentiate RISC and CISC architecture.	06	CO2	BTL-3
Q.3	**Answer the following: (Any Two)**	**12**		
1	What is microprocessor 8085? Explain the features of 8085 in detail.	06	CO4	BTL-4
2	What is array processor? Explain the categories of array processor.	06	CO3	BTL-4

3	Write the flag register and explain each of the flag with an example.	06	CO4	BTL-5
Q.4	**Answer the following: (Any One)**	**06**		
1	Which are the different phases of Instruction Cycle? Describe Register transfer for fetch phase with its diagram.	06	CO5	BTL-5
2	Explain the different types instructions with examples.	06	CO5	BTL-4

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT(DS) Sem-II Regular April-June 2024

U72A5BS-BASIC STATISTICS

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern:BTL1-Remember; BTL2- Understand; BTL3-Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	The marks out of 60 obtained by 54 students in a certain examination are given below. Calculate the Median marks.	06	CO1	BTL5

Marks	15-20	20-25	25-30	30-35	35-40	40-45	45-50	50-55
No. of student	4	5	11	6	5	8	9	6

2	The table below shows the number of men in various age group with some form of paid employment in a village. Find the modal age.	06	CO1	BTL5

Age :	10-20	20-30	30-40	40-50	50-60	60-70	70-80
f :	12	14	26	35	23	5	1

3	Find Q_1 and Q_2 for the following data.	06	CO1	BTL4,5

Marks	:20-30	30-40	40-50	50-60	60-70	70-80
No. of student	: 10	20	15	10	20	5

4	Find the standard deviation for the following data.	06	CO1	BTL2,5

Age	: 0-5	5-10	10-15	15-20	20-25	25-30
No. of Accidents	: 15	40	30	20	15	20

5 Find Karl Pearson's co-efficient of correlation for the following data. 06 CO2 BTL4,5

X	12	14	16	18	20	22	24
Y	12	10	8	6	4	2	0

6 Find Regression coefficient of marks in mathematics on marks in hindi for following bivariate frequency distribution. 06 CO3 BTL4,6

Marks in Hindi	Marks in Mathematics			
	5-15	15-25	25-35	35-45
0-10	1	1	0	0
10-20	3	6	5	1
20-30	1	8	9	2
30-40	0	3	9	3

7 A number is selected from the natural numbers less than 50. What is the probability of 06 CO4 BTL5

(a) Getting an even number?

(b) Getting a multiple of 10?

(c) Getting a perfect square?

(d) Getting an odd number?

Section – II

Q-2 Answer the following (Any Two)

		Marks	CO	Bloom's Taxonomy
		12		
1	An agency interviewed 200 Person as part of an opinion poll. The following distribution represents the age of person interviewed. Calculate the average age.	06	CO1	BTL2,3

Age :	80-89	70-79	60-69	50-59	40-49	30-39
f :	2	2	6	20	56	40

2	Find Geometric mean for the following data.	06	CO1	BTL2,3

Class :	10-20	20-30	30-40	40-50	50-60
f :	2	5	7	8	10

3	Find the Co-efficient of Variance for the following data.	06	CO1	BTL5

Class	0-5	5-10	10-15	15-20	20-25
f	10	20	5	3	10

Q.3 Answer the following: (Any Two)

		12	CO	BTL
1	Find Karl Pearson's co-efficient of correlation form the following bivariate frequency distribution.	06	CO2	BTL6

Weight (in-pounds)	Height (in inches)			
	40-44	44-48	48-52	52-56
35-55	4	40	60	--
55-75	---	---	24	88
75-95	---	2	---	8
95-115	3	--	---	--

2	Find Co-efficient of correlation and probable error for the following data.	06	CO2	BTL2,5

X	30	40	60	20	50
Y	4	8	6	4	3

3	Find Regression line of X on Y for the following data. Also estimate the value of X for Y=30.	06	CO3	BTL3,4

X	30	40	50	60
Y	10	15	17	20

Q.4	**Answer the following: (Any One)**	06	CO	BTL
1	One bag has 4 white balls and 3 black balls and a second bag contains 3 white balls and 5 black balls. One ball is drawn from first bag and placed without noticing the colour in the second bag. What is the probability that a ball now drawn from second bag is black?	06	CO4	BTL5
2	In a factory which manufactures bolts, machine A, B and C manufacturers respectively 25%, 35% and 40% of the bolts. Of their outputs 5, 4 and 2 per cent are respectively defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability it is manufactured by the machine A?	06	CO4	BTL5

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc IT (DS) Sem-II Regular April-June 2024

U72B6CS2: LANGUAGE AND COMMUNICATION SKILLS – II

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

		Marks	CO	Bloom's Taxonomy
Q-1	**Answer the following. (Any 5)**	**30**		
1	Write expressions for the following situations: a) Asking for a help b) Inviting a friend for a college farewell party c) Expressing gratitude for receiving the prize in a competition	06	CO2	2,3,6
2	What are the modals? Explain any four with example.	06	CO1	1,3,6
3	What is a presentation? What are the components of a presentation?	06	CO3	1,2
4	What is an advertisement? Explain the various types of advertisements with example.	06	CO4	1,2,4
5	What is a notice writing? Explain the various parts of it.	06	CO4	1,3
6	Explain the equation E=A+I in terms of letter writing.	06	CO4	2,4
7	What is agenda writing? Explain the steps in preparing a good agenda.	06	CO4	1,3,4

Section – II

		Marks	CO	Bloom's Taxonomy
Q-2	**Answer the following (Any Two)**	**12**		
1	**Do As Directed:** **(I) Use the suffix and make a word: '-ity'** **(II) Use the prefix and make a word: 'il-'**	06	CO1	2,3,4,5

	(III) **Give Synonyms for the word:** 'Prepared'			
	(IV) **Give Antonyms for the word:** 'Up stairs'			
	(V) **Give one word for:** "One who does not eat meat."			
	(VI) **Explain the difference between:** 'Accept' and 'except'	06	CO1	2,3,4,5
2	**Do As Directed:**			
	(I) Seema is clever. She____**(can/may)** answer difficult questions.			
	(II) None of my relatives ___ **(have/has)** been to the UK before.			
	(III) Identify the type of structural type of sentence : Though he was ill, he went to college.			
	(IV) You are still in your _____ **(adolescent/adolescence)**			
	(V) _____ **(would / can)** you please share your notes with me?			
	(VI) If Meena won the lottery, she_____ a big house. **(will buy, would buy, would have bought)**	06	CO4	1,2,3,6
3	Make an announcement regarding the workshop that your department is going to organize on the topic, 'Use of Data in Industry'. You can include details like, date, time, location, registration, contact information and prize, etc.			
Q.3	**Answer the following: (Any Two)**	**12**		
1	Shruti Singh drafts an email to her class teacher requesting her to allow the late submission of her project. Draft the email on her behalf.	06	CO4	1,2,3,6
2	Explain the mental steps in preparing a presentation.	06	CO4	1,3
3	As a student secretary, draft an agenda and MoM of the meeting held to discuss the first year students welcome party held at your campus.	06	CO4	1,2,3,6
Q.4	**Answer the following: (Any One)**	**06**		
1	What are the Dos and Don'ts of a GD?	06	CO5	1,3,5
2	What is a GD? What are the prerequisites of a GD?	06	CO5	1,2,3

**** End of the Paper ****